

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech.

Branch: Mechanical Engg./ Mechanical Engg. (Sandwich)

Semester : IV

Subject Code & Name: BTMPE405C Fluid Machinery

Max.Marks: 60

Date: 31/12/2024

Duration: 2 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 Which of these statements hold true?	(CO1)	1
a. Momentum conservation is applicable to neither individual directions nor the whole system		
b. Momentum conservation is applicable to the whole system but not individually		
c. Momentum conservation is applicable to both individual directions and the whole system		
d. Momentum conservation is applicable only to the three directions individually		
2 Force exerted by a jet on a moving plate happens in how many cases?	(CO1)	1
a. 3 cases b. 2 cases c. 1 case d. Nil		
3 The force analysis on a curved vane is understood using_____	(CO4)	1
a. Velocity triangles b) Angle of the plate		
c) Vane angles d) Plate dimensions		
4 The physical property Φ of the general transport equation is replaced by _____ to get momentum equation	(CO4)	1
a. Velocity vector b) Mass		
c) Force vector d) Acceleration vector		
5 Which among the following is the formula for Force when it strikes the plate?	(CO1)	1
a. $\rho a v^2$ b. $\rho a v$ c. ρa d. $\rho a E$		

21297378	6	Which among the following is true for hydroelectric power plants?	(CO2)	1
		a. Operating cost is low and initial cost is high		
		b. Both operating and initial cost are high		
		c. Both operating and initial cost are low		
		d. Operating cost is high and initial cost is low		
21297378	7	Centrifugal pumps transfer energy from _____	(CO 6)	1
		a) Rotor to fluid		
		b) Fluid to rotor		
		c) Draft to rotor		
		d) Rotor to draft		
21297378	8	Centrifugal pumps transport fluids by converting _____	(CO 6)	1
		a. Kinetic energy to hydrodynamic energy		
		b. Hydrodynamic energy to kinetic energy		
		c. Mechanical energy to kinetic energy		
		d. Mechanical energy to Hydrodynamic energy		
21297378	9	What is the primary objective of turbine governing?	(CO 5)	1
		a. Maintain constant expansion		
		b. Maintain constant pressure		
		c. Maintain constant speed		
		d. Maintain constant temperature		
21297378	10	Buckets and blades used in a turbine are used to:	(CO 2)	1
		a. Switch off the turbine		
		b. To regulate the wind speed		
		c. Alter the direction of water		
		d. To regenerate the power		
21297378	11	The discharge capacity of the reciprocating pump is _____ that of the centrifugal pump.	(CO 6)	1
		a. higher than		
		b. lower than		
		c. same as		
		d. unpredictable		
21297378	12	Which of the following pumps is used to pump sewage solids with liquid sewage without clogging the pump is:	(CO 6)	1
		a. Centrifugal Pump		
		b. Pneumatic Ejector		
		c. Reciprocating Pump		
		d. none of these.		
Q.2		Solve the following.		12
A)		Derive the equation for force exerted by the jet on moving plates with neat sketch.	(CO 4)	6
B)		Explain impulse-momentum principle with neat sketch.	(CO 1)	6

Q.3	Solve the following.	12
A)	Explain the phenomenon of Cavitation. How it can be prevented?	(CO 2) 6
B)	Draw the velocity triangle and derive the equation for workdone in Pelton Turbine.	(CO 4) 6
Q.4	Solve Any Two of the following.	12
	Explain the construction and working principle of Pelton Wheel with a neat sketch.	(CO 3) 6
	Explain the significance of Draft Tube with neat sketch.	(CO 2) 6
	A Kaplan turbine produces 60000 kW under a net head of 25 m with an overall efficiency of 90%. Taking the value of speed ratio K_u as 1.6, flow ratio ψ as 0.5 and the hub diameter as 0.35 times the outer diameter, find the diameter and speed of the turbine.	(CO 4) 6
Q.5	Solve Any Two of the following.	12
A)	What do you understand by governing of hydraulic turbines? Explain the working of a Governor with neat sketch.	(CO 4) 6
	Explain various types of controllers in a governor with neat sketch.	(CO 5) 6
	Explain the different types of casings used in Centrifugal Pump with neat sketches.	(CO 7) 6
Q.6	Solve Any Two of the following.	12
A)	A centrifugal pump has the following characteristics: outer diameter of impeller = 800 mm; width of impeller vanes at outlet = 100 mm; angle of impeller vanes at outlet = 40° . The impeller runs at 550 RPM and delivers 0.98 cubic metres of water per second under an effective head of 35 m. A 500 kW motor is used to drive the pump. Determine the manometric, mechanical and overall efficiencies of the pump. Assume water enters the impeller vanes radially at inlet.	(CO 7) 6
	What do you understand by special purpose pumps? Enlist and explain the conditions in which these are preferred?	(CO 6) 6
	Explain principle, construction and working of Reciprocating and Gear Pump with neat sketches.	(CO 6) 6
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